Yanyu Chen

Robotics | Software · Mechanical

Contact Information

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Portfolio: https://yanyu-chen.com/

Summary

Year 2 NUS Robotics and Machine Intelligence undergraduate. As sole engineer at MUJIN, built and deployed an end-to-end dual-arm system based on a diffusion policy on a humanoid robot, covering teleoperation, data collection, real-time control and safety. Member of the Mechanical Team of the NUS DJI RoboMaster team and Captain of the RMUL division this year. Working proficiency in English (Fluent), Chinese (Native) and German (TELC C1).

Education

National University of Singapore (NUS)
Robotics and Machine Intelligence (B.Eng) | Minor: Innovation and Design
Aug 2025 – May 2029 (expected) | GPA: 4.63/5.00 | Singapore

Frankfurt International School
International Baccalaureate (IB Diploma)
Aug 2019 – May 2025 | Honours: 2023-2025 highest distinction, Physics Award (Class of 2025) |
IB Score: 43/45 (Top 2% Global) | Frankfurt, Germany

Technical Skills

Control & Algorithms | PyTorch, imitation learning (diffusion policy), OpenCV, ROS2, Gazebo, LiDAR/vision integration

Mechanical Design | SolidWorks, Fusion 360, FEA, DFM/DFA; 5+ years CAD and engineering drawing

Embedded Systems | STM32 HAL/LL, Jetson Orin (CUDA), PID, motor control, CAN/RS485

Coding and Tools | Python, C++, CMake, Git, STM32CubeIDE

Experiences

MUJIN Corp.

Software Intern, Humanoid Robotics R&D | May – Jul 2026 | Guangzhou, China

10-week proof of concept: imitation-learned dual-arm handling of soft parcels for a warehouse sorting scenario on an AgiBot Genie G1 humanoid.

Full report, code and engineering documentation: https://github.com/yammmyu/Intern_Summer_2026

- Single-handedly delivered a fully autonomous "grasp → lift → label-check → flip → place" cycle for a warehouse sorting scenario.
- Built in self-recovery from failed grasps, with the model running end to end on the robot's own three cameras. **~90% task success** at **~12 s median cycle** over 38 scored trials.
- Built from scratch the integration layer from the **10 Hz** diffusion policy's predictions to a **120 Hz** arm controller, using time-uniform substep expansion, cross-chunk temporal ensembling and Gaussian trajectory smoothing. Replaced clock-based synchronisation with **absolute row IDs** keyed to the robot's own execution clock, eliminating the motion oscillation caused by varying inference times.
- Designed a layered safety envelope of **11 named invariants** (latched E-stop, per-command displacement watchdog, workspace bounds, mandatory sim validation before any command reaches the robot), enforced by a **pytest suite run as a launch pre-flight** that blocks startup on regression.

- Completed the handover and wrote the bilingual (EN/CN) technical documentation and proof-of-concept report.

NUS Calibur Robotics (DJI RoboMaster)

Mechanical Team— "Engineer" robot 6-DOF arm | Aug 2025 – Apr 2026

- Designed the Engineer robot's 6-DOF arm; my part (the Engineer Challenge) won **First Prize**:
  - 2026 RoboMaster University League (Chongqing) Engineer Challenge, **2nd place.**
- Cut **fasteners by 40%** and **weight by 25%** against the previous generation through truss topology optimization and interlocking parts.
- Designed an internal cable-routing system with conductive slip rings, protecting the wiring and allowing 360° continuous rotation of the end effector.
- Applied design-for-assembly throughout: quick-release structures let any component be replaced in under 3 minutes.

Captain, Competition Team — the team's RMUL-focused division | May 2026 – Present

- Lead the division: set the season phase map, plan projects and assign tasks across members, and own supplier contact and inventory.
- Building a portable moving target, so auto-aim can be tested without tying up a second robot and driver. Planned with a product requirements document and 26-week phased roadmap **(~US$910 budget).**

Projects

TurtleBot3 Simulated Warehouse Transportation Project

Coursework (CDE2310) | Jan – Apr 2026 | Mechanical + Software

- Owned the **mechanical design** (9-ball gravity-fed magazine and cam launcher within a 5 cm height budget).
- Owned **ArUco marker-based docking**: pose estimation that resolves the two-solution ambiguity of square markers, plus a one-shot docking plan.

Ybot - Desktop 6-DOF Robotic Arm

Personal Project | Jun 2023 – Aug 2025 | Mechanical + Firmware

- Started from a market gap: precise arms cost thousands of dollars and hobby kits are imprecise. Planned the whole system, then **designed and built the arm at a unit cost of S$611** (Fusion 360, harmonic drives, 3D-printed PETG) and wrote a C++ ROS 2 node to drive it from MoveIt 2 plans.

Awards & Certifications

2026 RoboMaster University League (Chongqing)| Engineer Challenge| 2nd Place / 1st Prize

HIMCM Mathematical Modeling Competition | M Award (Top 4 in Europe, 2023)

SolidWorks CSWP (Full Score)

Languages

English: Fluent (academic / professional) **Chinese: Native** **German: Proficient** (TELC C1)